



DWH

-An Overview

INTRODUCTION

BUSINESS PERSPECTIVE

➤ **From the business perspective, the requirements of the enterprise fall into categories illustrated and described below:**

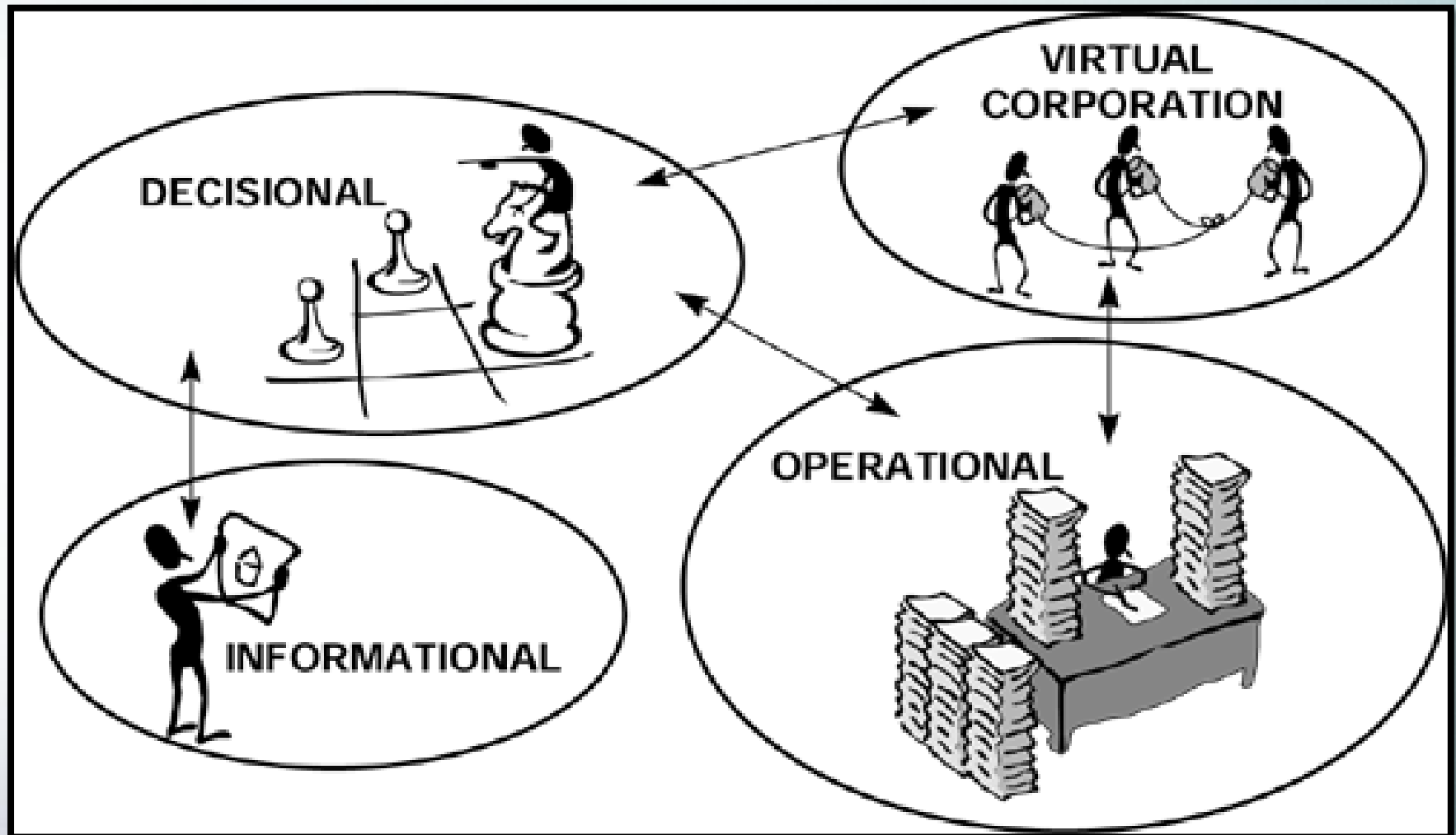
➤ **OPERATIONAL**

➤ **INFORMATIONAL**

➤ **DECISIONAL ' & '**

➤ **VIRTUAL CORPORATION**

VIEW IN BUSINESS PERSPECTIVE



Operational

- **Technology supports the smooth execution and continuous improvement of day-to-day operations.**
- **The identification and correction of errors through exception reporting and workflow management, and the overall monitoring of operations.**
- **Information retrieved about the business from an operational viewpoint is used to either complete or optimize the execution of a business process.**

Operational

➤ **Example:**

- Day-to-day transaction includes - ATM Withdrawals
which includes Checking Balance, Mini Statement,
Withdrawal, etc...

Informational

- **Technology** makes current, relatively static information widely and readily available to as many people as need access to it.
- **Examples** include company policies, product and service information, organizational setup, office location, corporate forms, training materials, company profiles.

Decisional

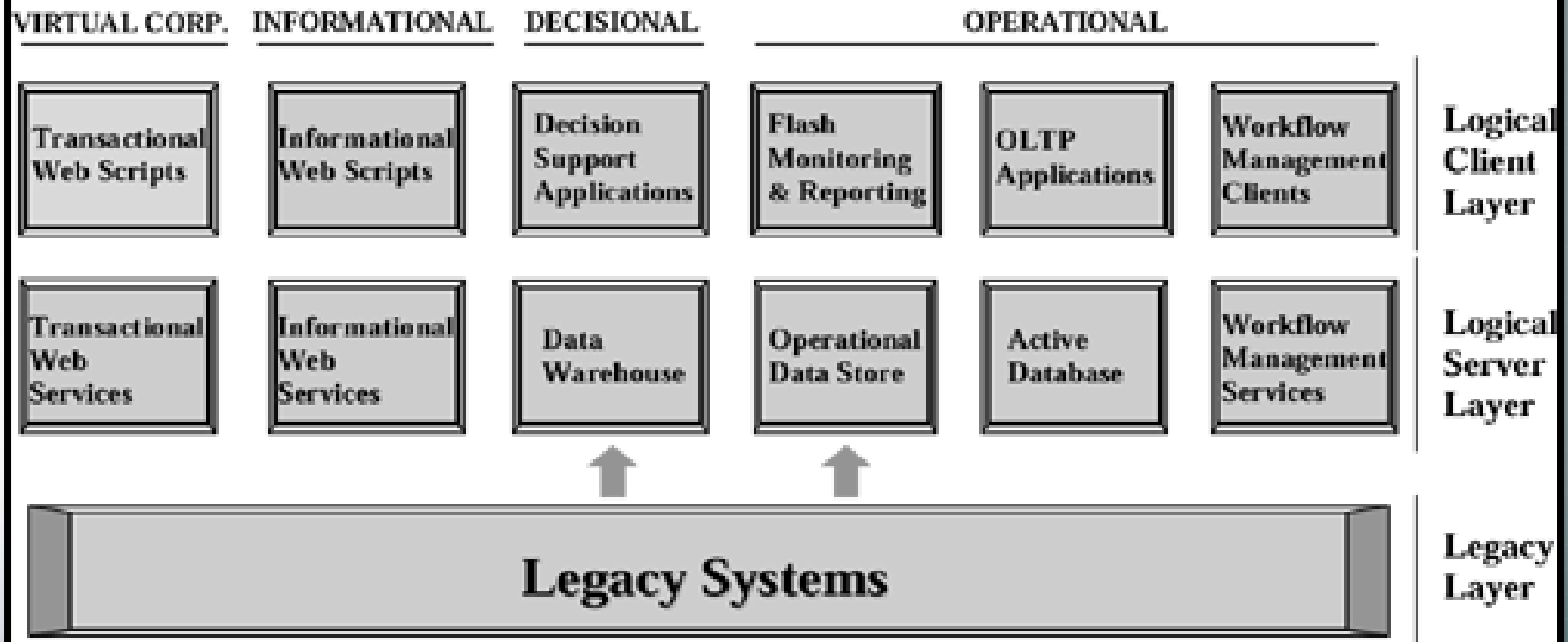
- **Technology supports managerial decision-making and long-term planning.**
- **Decision-makers are provided with views of enterprise data from multiple dimensions and in varying levels of detail. Historical patterns in sales and other customer behaviour are analysed.**
- **Examples** of Decisional systems support includes decision-making and planning through scenario-based modelling, what-if analysis, trend analysis, and rule discovery.

Virtual Corporation

- **Technology enables the creation of strategic links with key suppliers and customers to better meet customer needs.**
- **In the past, such links were feasible only for large companies because of economies of scale.**
- **Now, the affordability of Internet technology provides any enterprise with this same capability.**

HIGH LEVEL ARCHITECTURE

INFOMOTION ENTERPRISE ARCHITECTURE



Business System

- **We can divide IT systems into transactional (OLTP) and analytical (OLAP).**
- **In general we can assume that OLTP systems provide source data to data warehouses, whereas OLAP systems help to analyze it.**

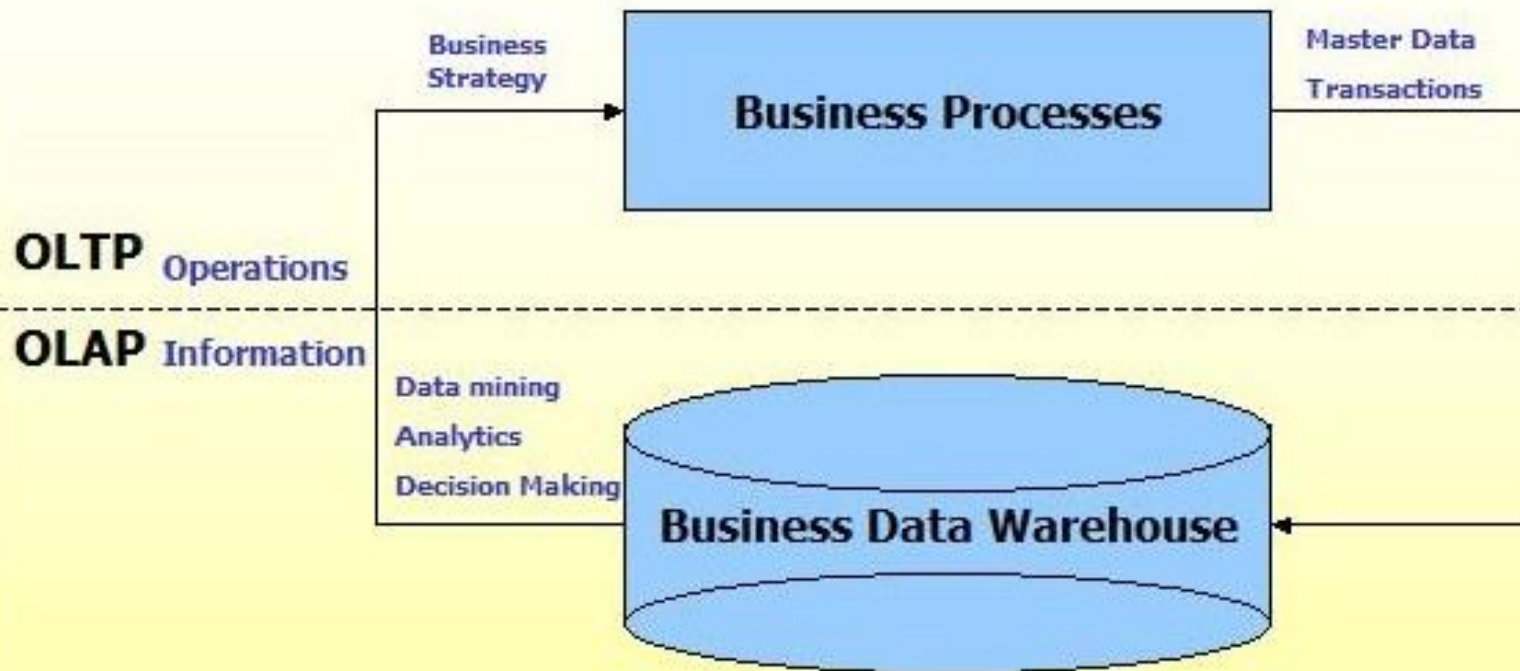
OLTP / Transactional Systems

- ***OLTP (On-line Transaction Processing)*** is characterized by a large number of short on-line transactions (INSERT, UPDATE, DELETE).
- The main emphasis for OLTP systems is put on very fast query processing, maintaining data integrity in multi-access environments and an effectiveness measured by number of transactions per second.
- In OLTP database there is detailed and current data, and schema used to store transactional databases is the entity model (usually 3NF).

OLAP / Analytical Systems

- ***OLAP (On-line Analytical Processing)*** is characterized by relatively low volume of transactions.
- Queries are often very complex and involve aggregations. For OLAP systems a response time is an effectiveness measure. OLAP applications are widely used by Data Mining techniques.
- In OLAP database there is aggregated, historical data, stored in multi-dimensional schemas (usually star schema).

OLTP (vs.) OLAP



Functions of Data Warehouse Tools and Utilities

- **The following are the functions of data warehouse tools and utilities:**
 - **Data Extraction** - Involves gathering data from multiple heterogeneous sources.
 - **Data Cleaning** - Involves finding and correcting the errors in data.
 - **Data Transformation** - Involves converting the data from legacy format to warehouse format.
 - **Data Loading** - Involves sorting, summarizing, consolidating, checking integrity, and building indices and partitions.
 - **Refreshing** - Involves updating from data sources to warehouse.

Differences Between OLTP and OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

DWH '&' BI

- An overview

What is Business Intelligence (BI)?

- The term BI was coined by Gartner group in 1993.
- It is an important component in today's business information systems environment.
- It is the process of turning data into knowledge and knowledge into business gains.
- It collects and stores data into meaningful information in order to achieve better and timelier business decisions.
- It is an end user's activity supported by various analytical and collaborative tools.

What is Business Intelligence (BI)?

- BI is used for enhancement and optimization of organizational performance and operation.
- It delivers critical business information to end-users.
- It supports internal enterprise users in the assessment.
- It is applied across disciplines, namely Finance, CRM, and SCM
- It encompasses all types of data such as RDBMS, text, hierarchical, audio, and video.

BI- Nutshell

- **Analyze** internal business activities to improve processes, increase efficiency, and reduce costs
- **Track** external market trends to understand customer behavior, improve relationships, identify opportunities, and increase competitiveness

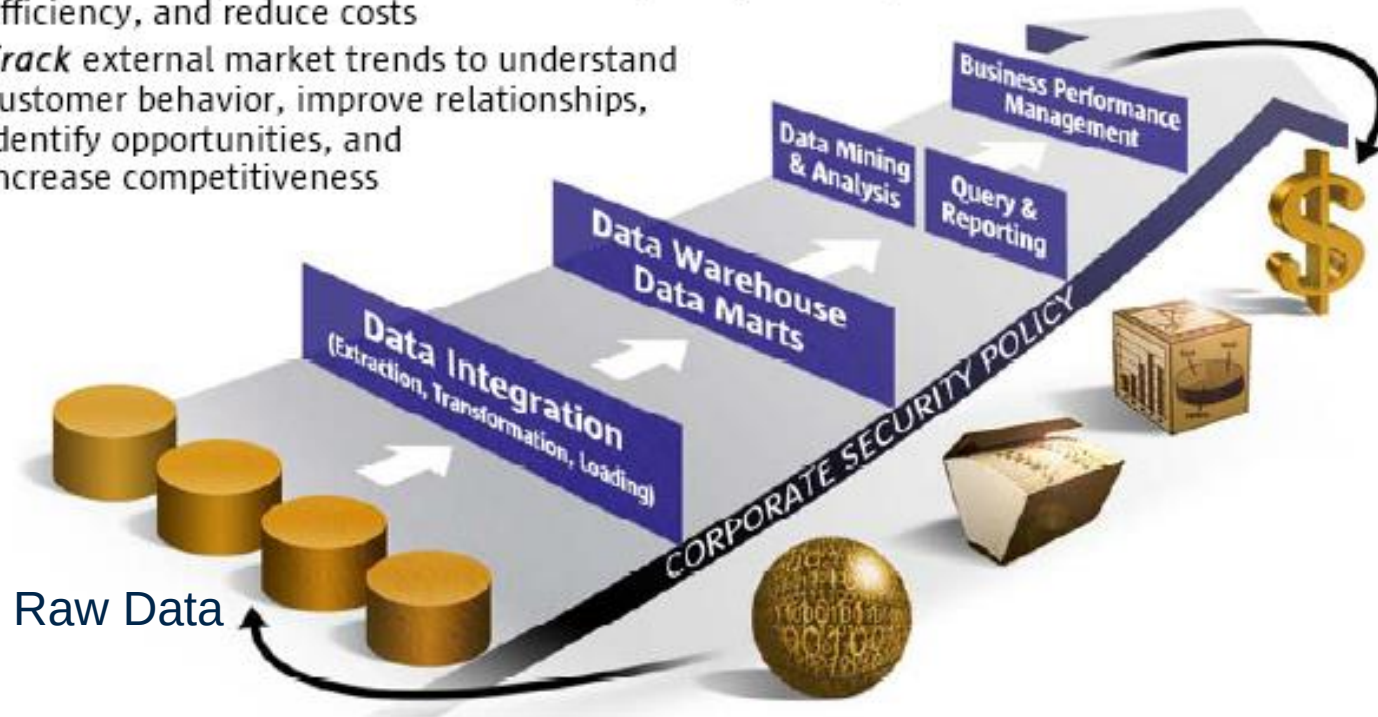


Figure 1. BIDW Overview

NEED FOR BI

PRODUCER WANTS TO KNOW

Which are our lowest/highest margin customers ?

Who are my customers and what products are they buying?

What is the most effective distribution channel?

What product promotions have the biggest impact on revenue?

Which customers are most likely to go to the competition ?

What impact will new products/services have on revenue and margins?



NEED FOR BI ...

- Data, Data everywhere yet ...
- I can't find the data I need
 - data is scattered over the network
 - many versions, subtle differences
- I can't get the data I need
 - need an expert to get the data
- I can't understand the data I found
 - available data poorly documented
- I can't use the data I found
 - results are unexpected
 - data needs to be transformed from one form to other



Why Business Intelligence?

- BI is required to meet the following business needs:
 - To support the process of exploring data, relationships existing within data, and trends
 - To make more accurate and more informed decision making
 - To provide timely and accurate information to better understand your organization and to make more informed, real-time decisions

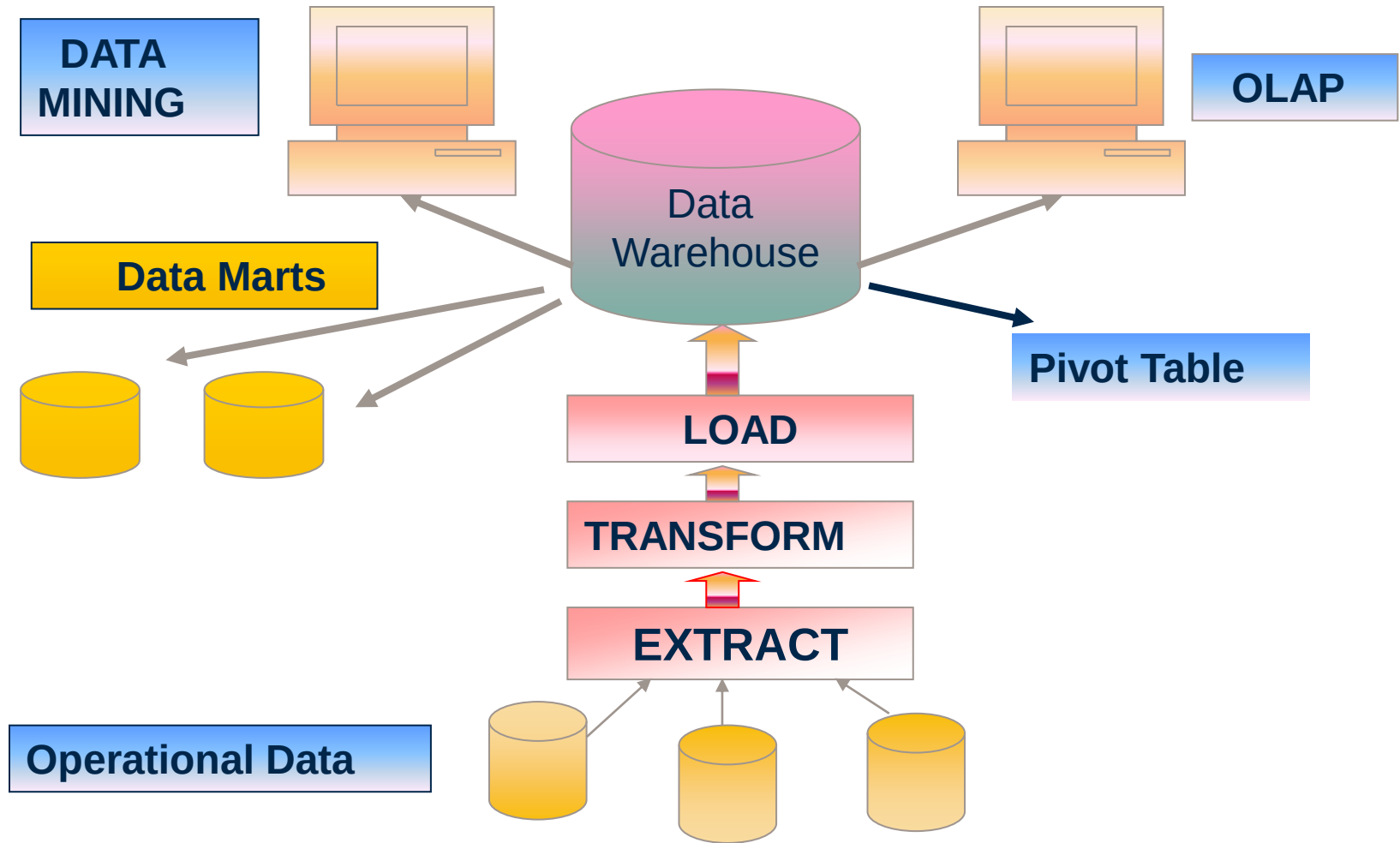
Why Business Intelligence?

- Data Analysis is a huge and crucial part of Business Intelligence.
- Many organizations need to know the overall performance and the way its business is functioning.
- BI is used to gather past as well as present data.
- Modern BI systems are capable of managing large amount of unstructured data.

Frequently used BI Terms

- Let us discuss some of the frequently used BI terms:
 - Relational Database (RDB)
 - Relational Database Management System (RDBMS)
 - Example: Informix, Microsoft SQL Server, Oracle.
 - Online Transaction Processing (OLTP)
 - Online Analytical Processing (OLAP)

Diagrammatic Representation



DWH Properties

“ A data warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of **management's decision making process.”**

➤ **The properties of DWH are as Follows:**

➤ **Subject-Oriented**

➤ **Integrated**

➤ **Non-Volatile**

➤ **Time Variant**

What is a Data Warehouse?

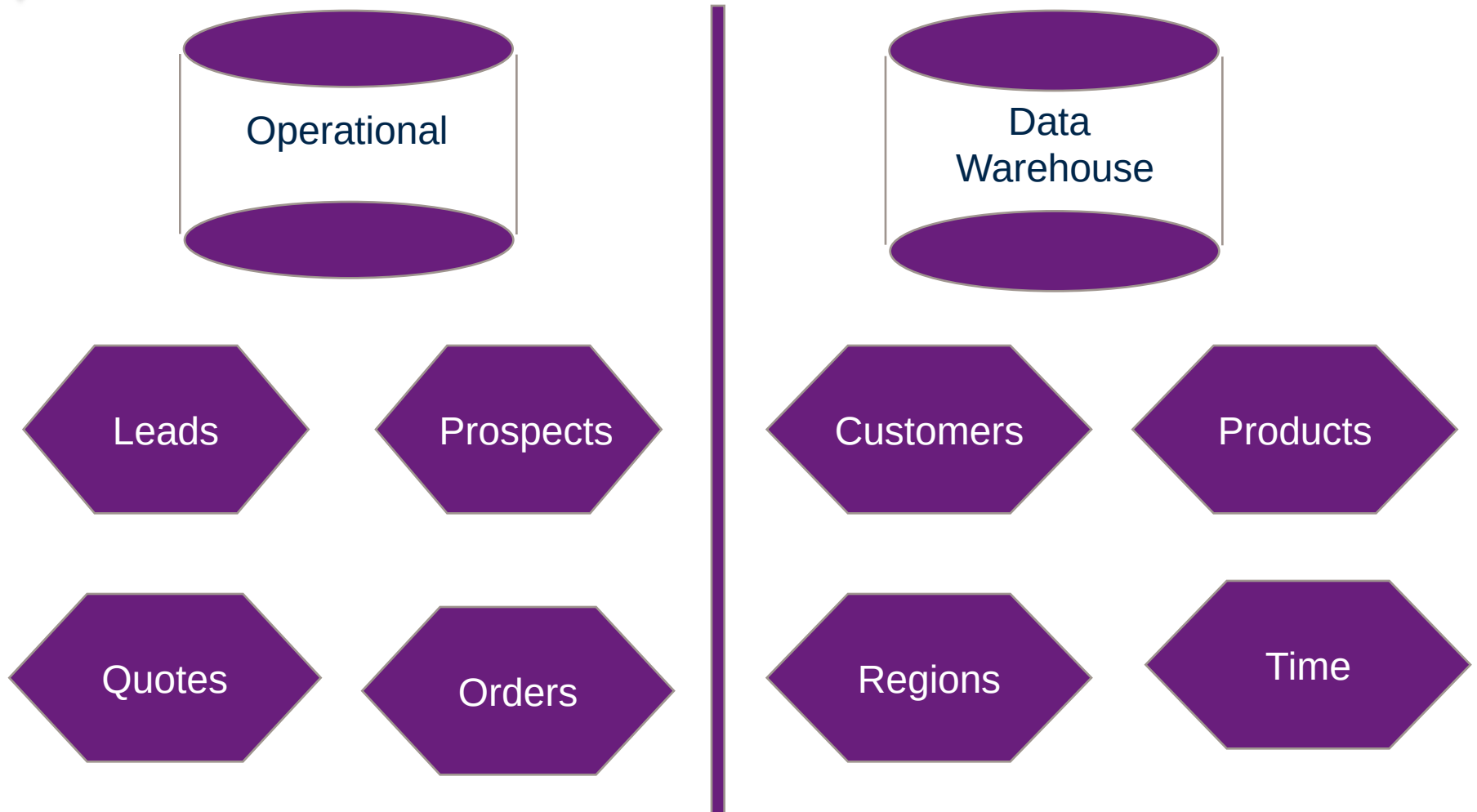
- Data Warehouse is a single, complete, and consistent store of data.
 - It is obtained from a variety of sources.
 - It is made available to users in a way they understand and use in a business context.
 - It is Central repository of information.
 - It is a collection of key information.
 - It contains read-only data.
 - It contains historical data used for analysis purpose.
 - It enables managers to make business decisions.

2.2 Characteristics of a Data Warehouse?

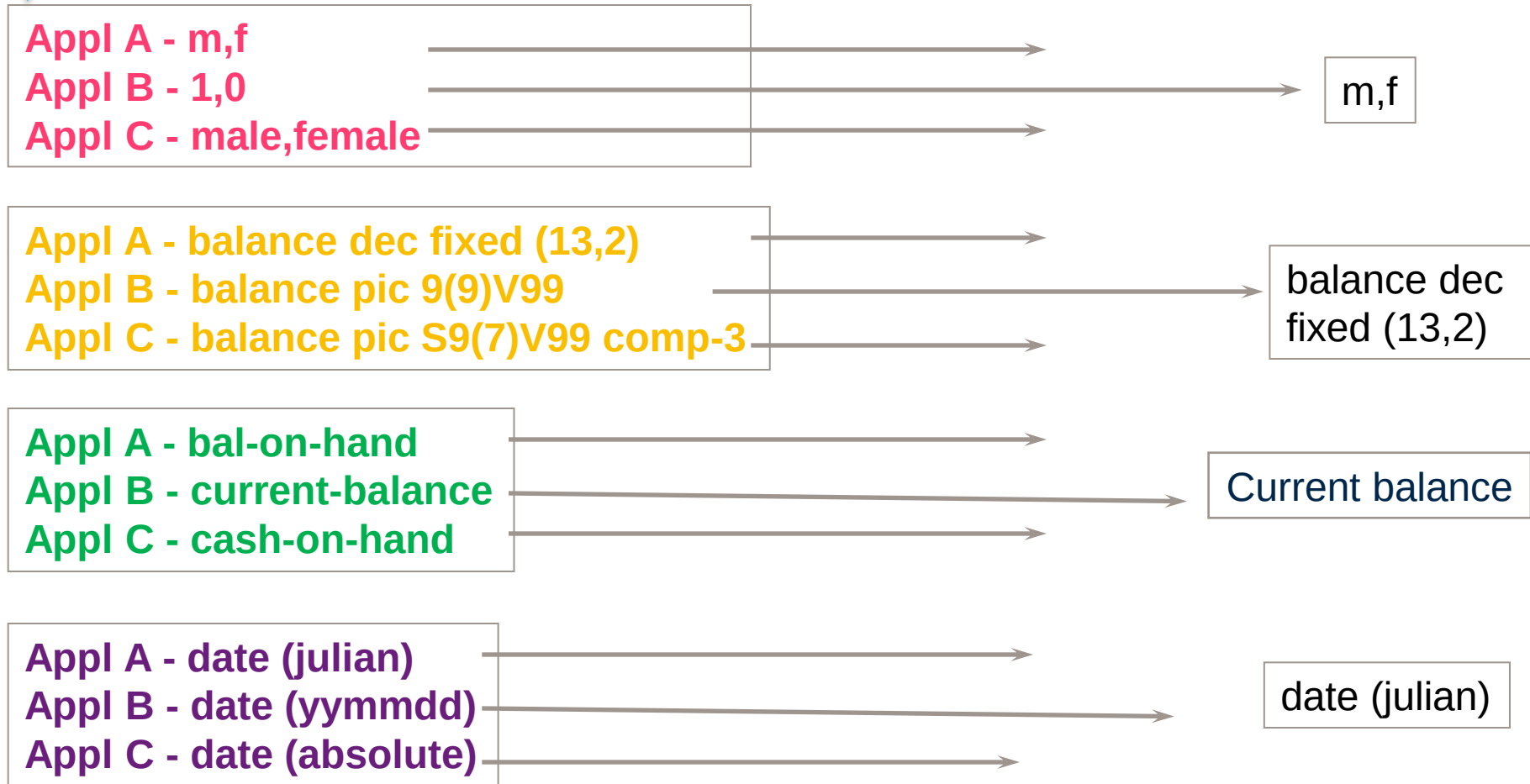
- “A data warehouse is a **subject-oriented, integrated, nonvolatile, time-variant** collection of data in support of management's decisions”.

- WH Inmon

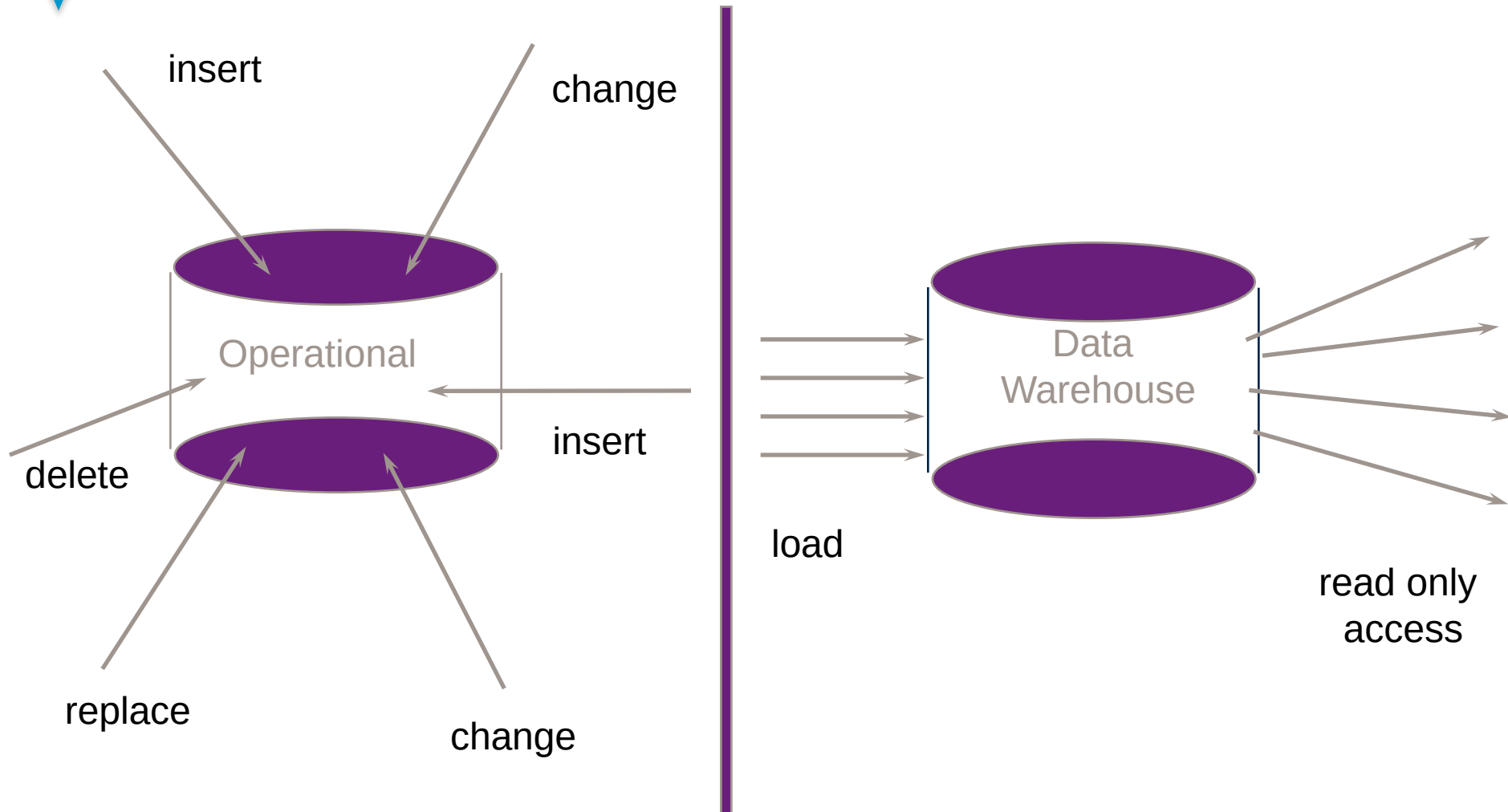
Subject-Oriented



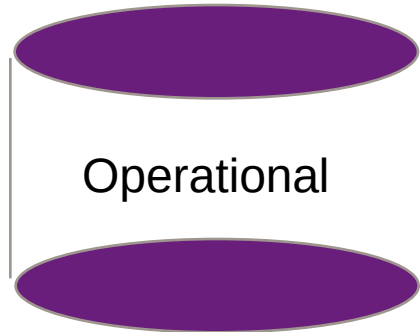
Integrated



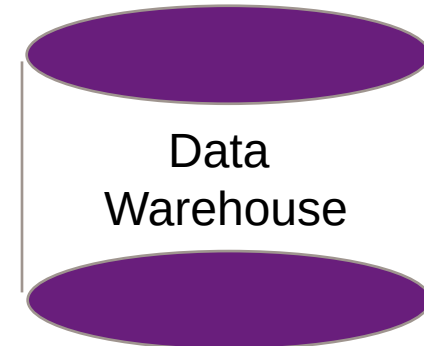
Non-volatile



Time Variant -



- **Current Value data**
- **time horizon : 60-90 days**
- **key may not have element of time**



- **Snapshot data**
- **time horizon : 5-10 years**
- **key has an element of time**
- **data warehouse stores historical data**

Evolution Of Data warehouse

- **60's: Batch reports**
 - hard to find and analyze information
 - inflexible and expensive, reprogram every new request
- **70's: Terminal-based DSS and EIS (executive information systems)**
 - still inflexible, not integrated with desktop tools
- **80's: Desktop data access and analysis tools**
 - query tools, spreadsheets, GUIs
 - easier to use, but only access operational databases
- **90's till now: Data warehousing with integrated OLAP engines and tools, real time DW**

Why Data Warehouse?

- Data Warehouse is required to meet the following needs:
 - Companies want to tap on the vast potential of information to:
 - Have a separate informational system from operational systems
 - Improve quality of decision making
 - Companies seek profitability through focused action.
 - IT business requires an integrated, company-wide view of high quality information.
 - Organizations want to analyze their activities in a balanced way.
 - Organizations need to build on Customer Relationship Management.

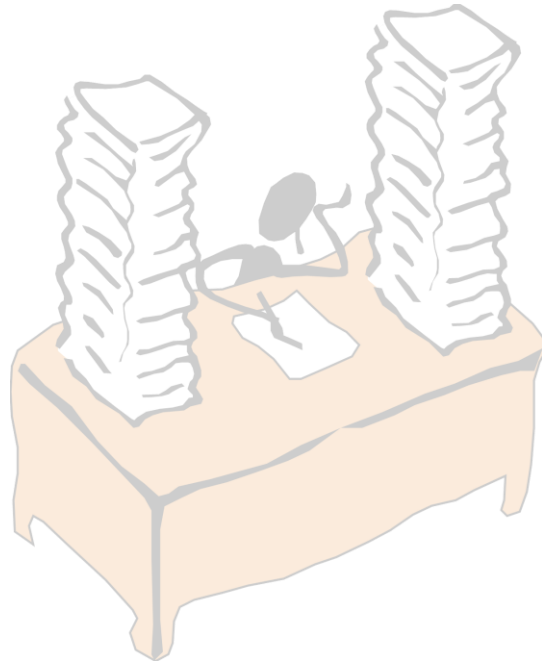
Why a separate Data Warehouse?

- A Data Warehouse helps in finding missing data.
- It provides consolidated data from multiple data sources.
- It helps in maintaining data quality coming from different sources.
- Special data organization is needed for vast volume of data.
- Complex OLAP queries degrade performance.

What is Data Warehouse Architecture?

- Data Warehouse Architecture is a description of the components and services of the Data Warehouse.
 - It provides the mechanism to achieve enterprise integration to support business.
 - It provides an organizing framework that will improve data sharing.

Data Mining works with Warehouse Data



- Data Mining provides the Enterprise with intelligence

- Data Warehousing provides the Enterprise with a memory



What makes data mining possible?

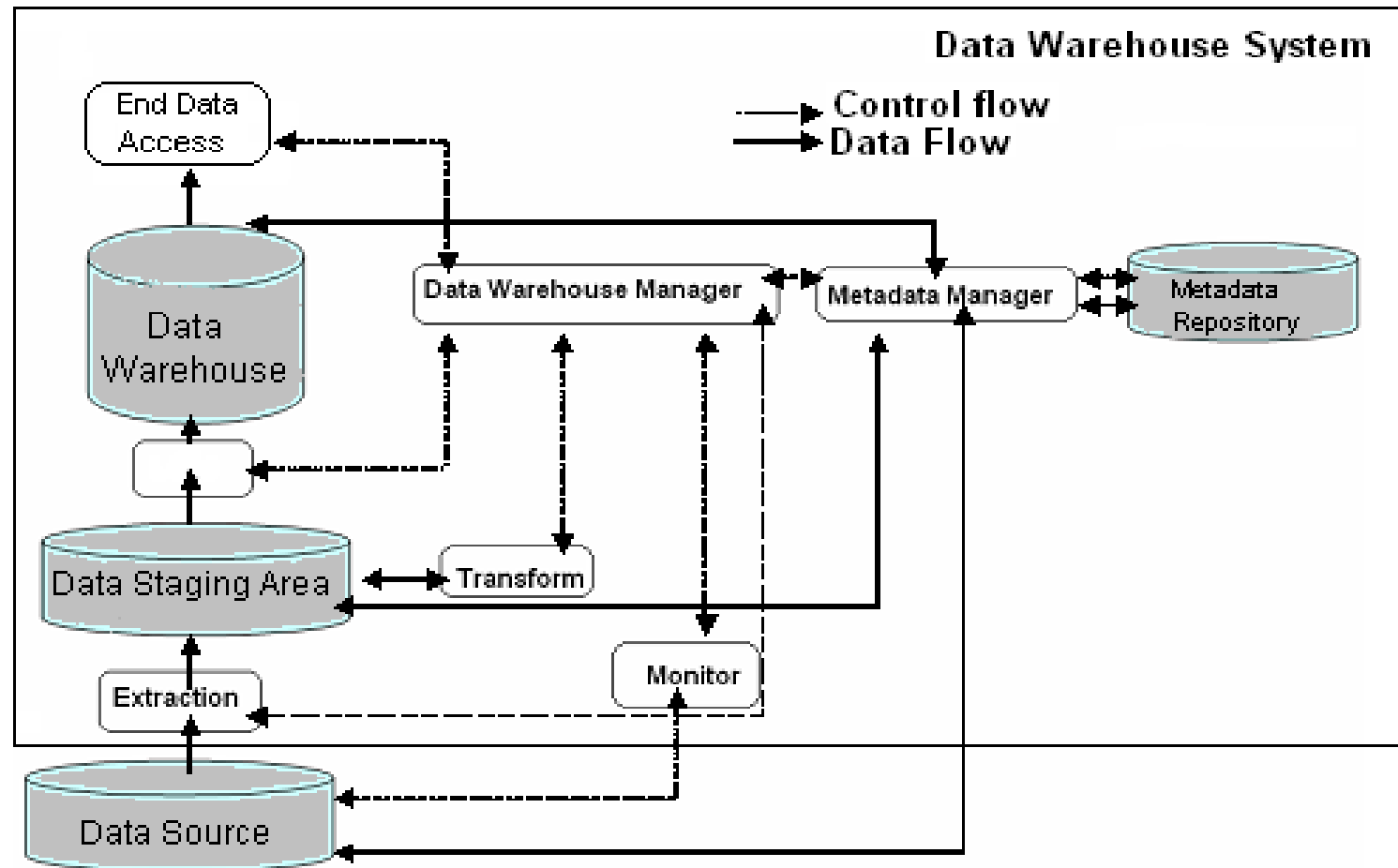
- Advances in the following areas are making data mining deployable:
 - Data warehousing
 - Better and more data (i.e., operational, behavioral, and demographic)
 - The emergence of easily deployed data mining tools and
 - The advent of new data mining techniques.

-- Gartner Group

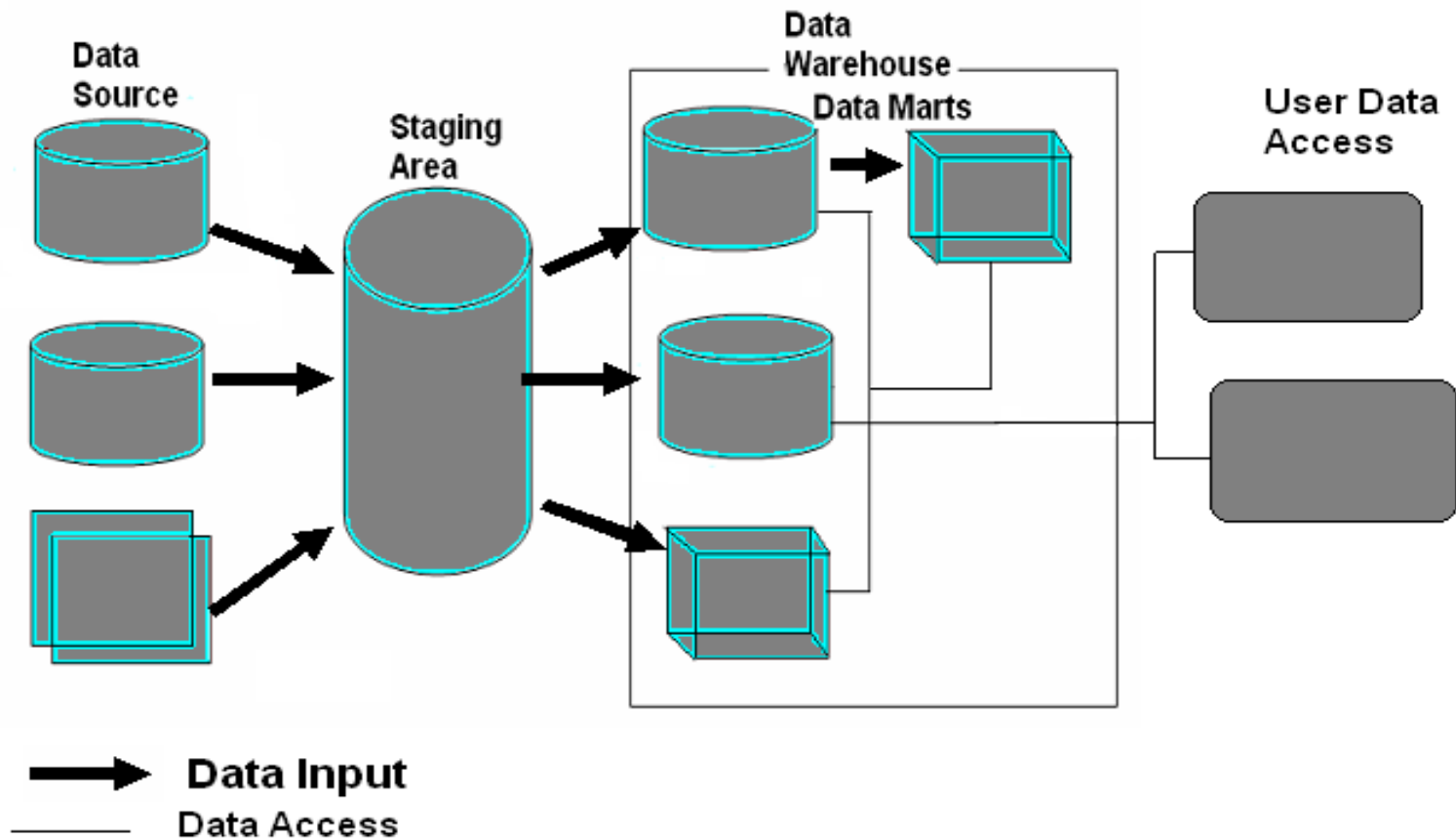
Data Warehouse Architecture Layers

- Data Warehouse Architecture consists of interrelated parts called as “layers” or “components”.
- Four layers of Data Warehouse Architecture are:
 - Operational: Functions as data storage
 - Informational: Stores business logic
 - Data access: Acts as a bridge between operational and informational layer
 - Meta data: Stores data dictionary

Block Diagram – Data Warehouse Architecture



Data Warehouse Components



Salient Features

- Here are some of the features of a Data Warehouse:
 - Time-variant data:
 - Data is meant for analysis and decision-making over the time.
 - Changes to the data are recorded against time dimension.
 - Data is stored as snapshots over past and current periods.
 - Non-volatile data:
 - Data is not needed to run the daily business.
 - Data is primarily used for query and analysis.
 - Individual transactions are not updated in a Data warehouse.
 - Data is never over-written or deleted. It is non-updatable data.

Salient Features

- Data granularity:
 - It refers to the level of detail.
 - It is inversely proportional to the amount of data stored.
 - Data is summarized at different levels.
 - Many Data warehouses have at least two levels of granularity.
 - Summarized data is stored.
 - It reduces storage costs.
 - It reduces CPU usage.
 - It increases performance since smaller number of records have to be processed.
 - Design is around traditional high level reporting needs.
 - Tradeoff with volume of data to be stored and detailed usage of data.

Salient Features

- Subject oriented:
 - Data is stored by subjects, not applications.
 - Data is organized in the Data Warehouse such that it will infer the real world.
 - Data is organized around major subjects, such as customer, product, sales.
 - Focus is on the modeling and analysis of data for decision makers.
 - DW provides a simple and concise view around a particular subject.
 - DW is organized around the key subject of the enterprise.
 - Major subjects may include customers, patients, students, products, and time.

Salient Features

- Integrated data:
 - Data is pulled from various databases from all applications.
 - Operational platforms and operating systems for the data could be different.
 - Data has to undergo a process of transformation, consolidation, and integration.
 - Data inconsistencies are removed, standardization is achieved.

What is a Data Mart?

- Data Mart is a subset of the Data warehouse.
 - It is typically fed from the Data warehouse.
 - It is a Data warehouse that has limited scope.
 - It is a repository of data gathered from operational data and other sources.
 - It is used for decision making by a particular end-user group.
 - Emphasis is on meeting the specific demands of a particular group of knowledge users.
 - Maintain the ability to access the underlying base data.

Types of Data Marts

- Dependent Data Mart

- A Data Mart whose source is the Data Warehouse
- All dependent Data Marts are loaded from the same source – the Data Warehouse

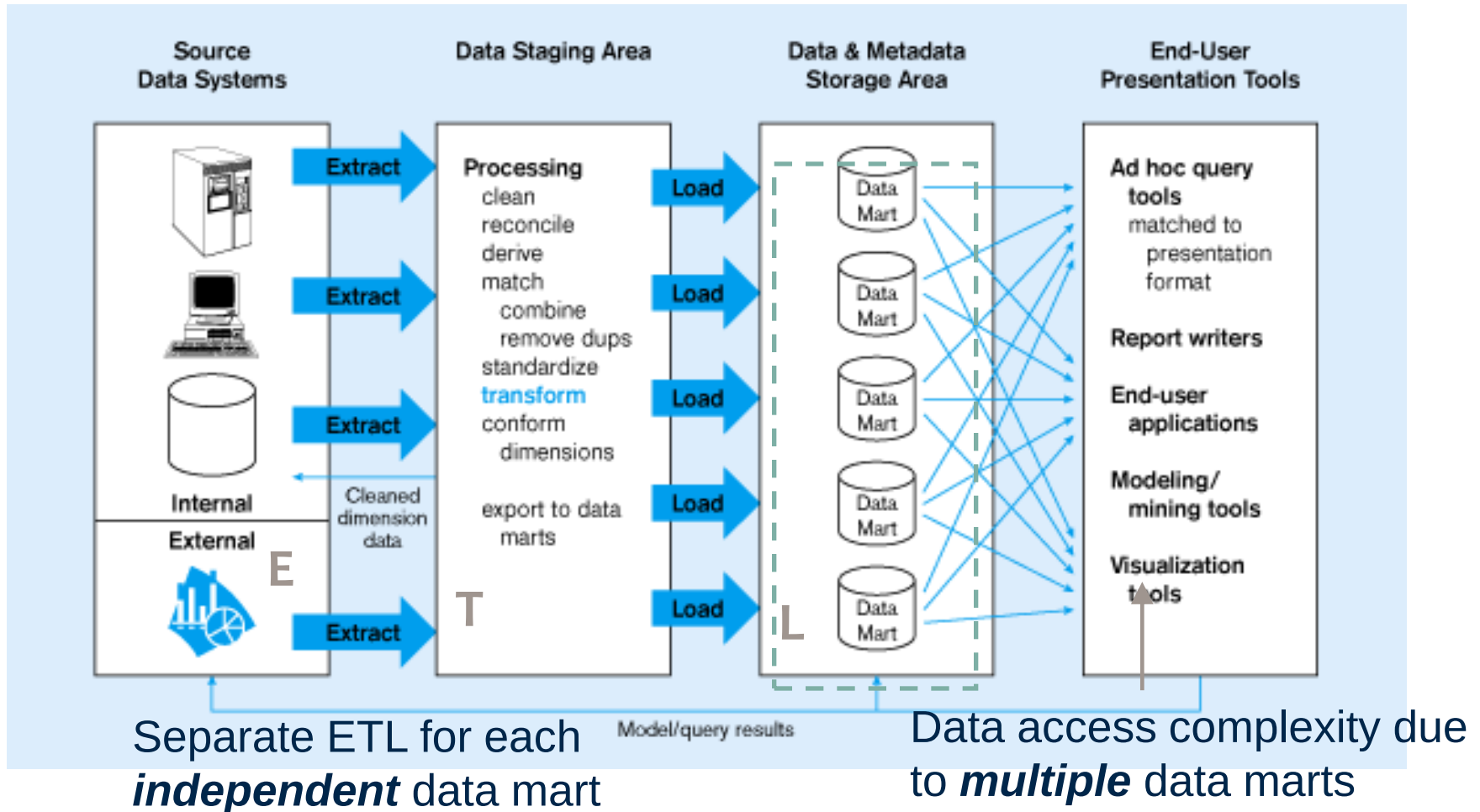
- Independent Data Mart

- A Data Mart whose source is the legacy application environment
- Each independent Data Mart is fed uniquely and separately by the individual source systems

Independent Data Mart

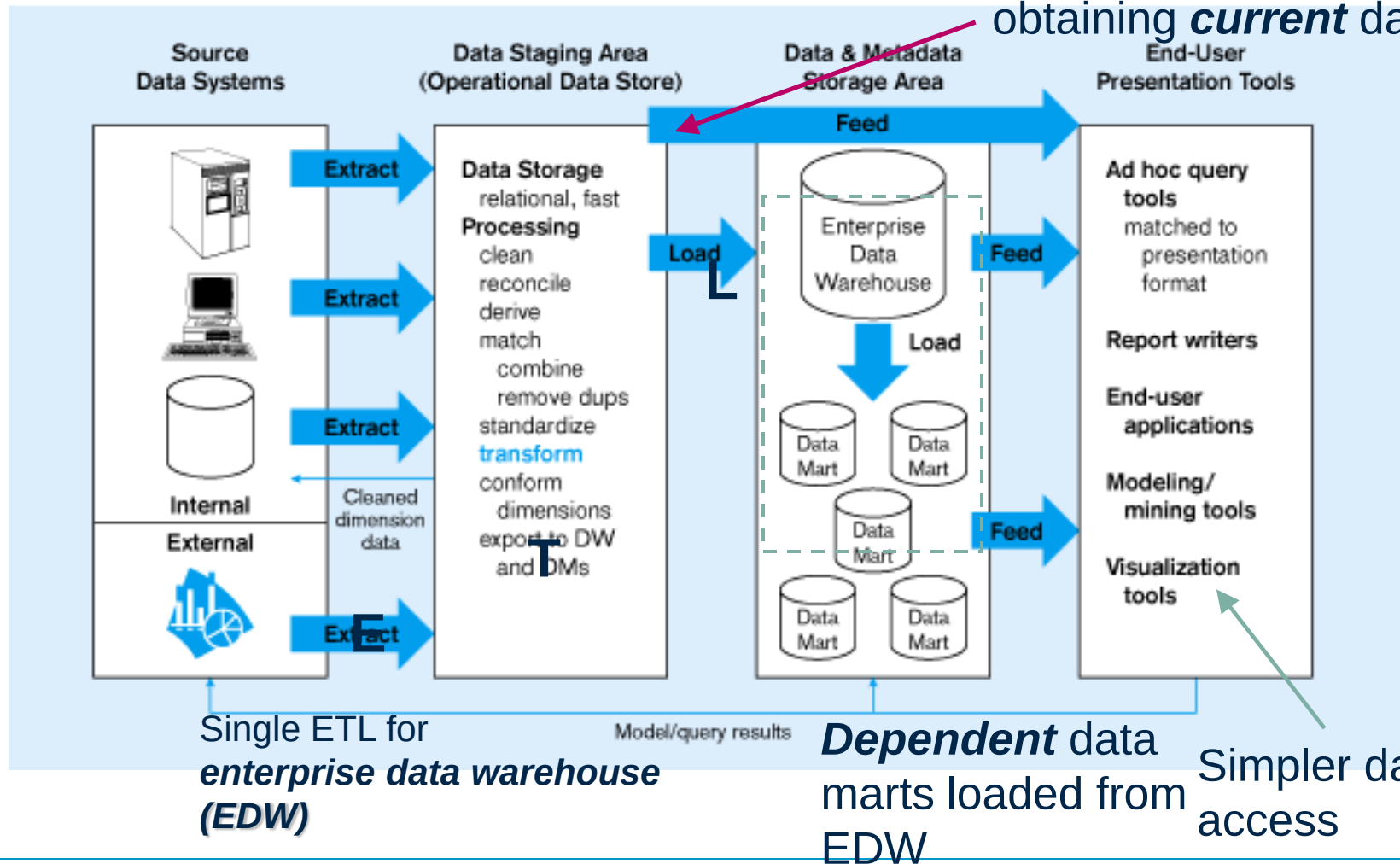
Data marts:

Mini-warehouses, limited in scope



Dependent data mart with operational data store

ODS provides option for obtaining **current** data



Industry-wise Application

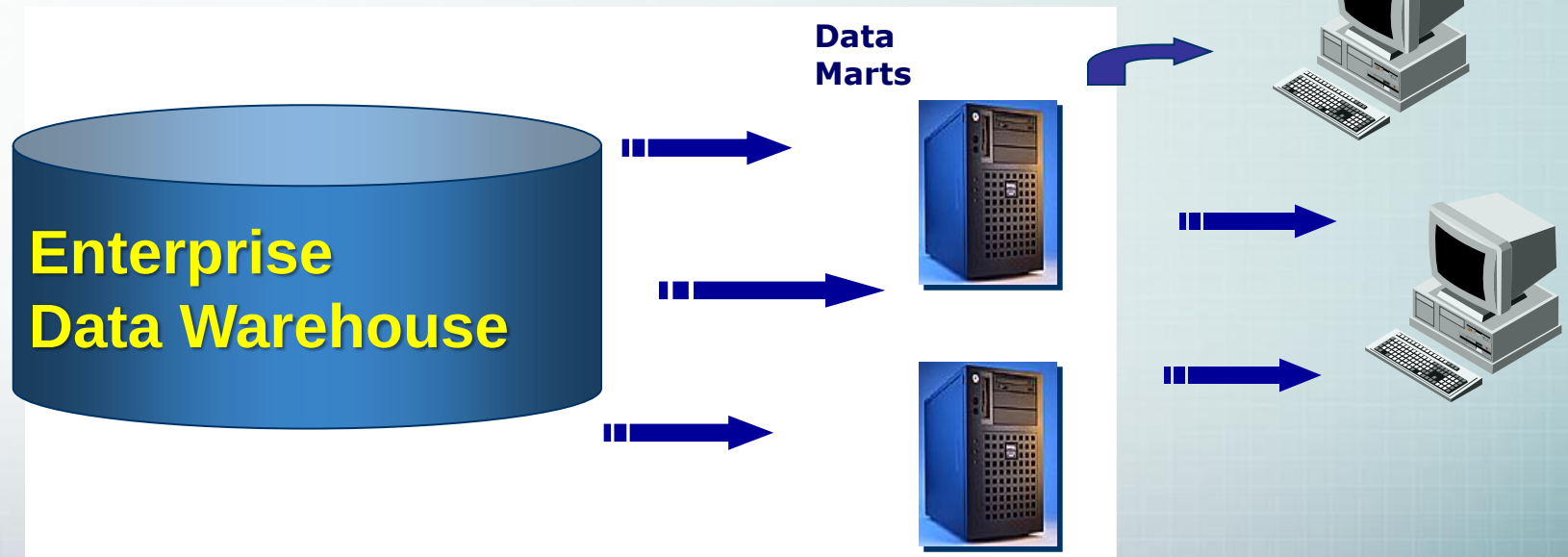
Industry	Application
Finance	Credit Card Analysis
Insurance	Claims, Fraud Analysis
Telecommunication	Call record analysis
Transport	Logistics management
Consumer goods	Promotion analysis
Data Service providers	Value added data
Utilities	Power usage analysis

DWH Layers

- **In general, all data warehouse systems have the following layers:**
 - **Data Source Layer**
 - **Data Extraction Layer**
 - **Staging Area**
 - **ETL Layer**
 - **Data Storage Layer**
 - **Data Logic Layer**
 - **Data Presentation Layer**
 - **Metadata Layer**
 - **System Operations Layer**

What is a Data mart?

- Data mart is a decentralized subset of data found either in a data warehouse or as a standalone subset designed to support the unique business unit requirements of a specific decision-support system.
- Data marts have specific business-related purposes such as measuring the impact of marketing promotions, or measuring and forecasting sales performance etc.,



Data marts - Main Features

Main Features:

- **Low cost**
- **Controlled locally rather than centrally, conferring power on the user group.**
- **Contain less information than the warehouse**
- **Rapid response**
- **Easily understood and navigated than an enterprise data warehouse.**
- **Within the range of divisional or departmental budgets**

Advantages of Datamart over Datawarehouse

- **Typically single subject area and fewer dimensions**
- **Limited feeds**
- **Very quick time to market (30-120 days to pilot)**
- **Quick impact on bottom line problems**
- **Focused user needs**
- **Limited scope**
- **Optimum model for DW construction**
- **Demonstrates ROI**
- **Allows prototyping**

DWH DESIGN APPROACHES

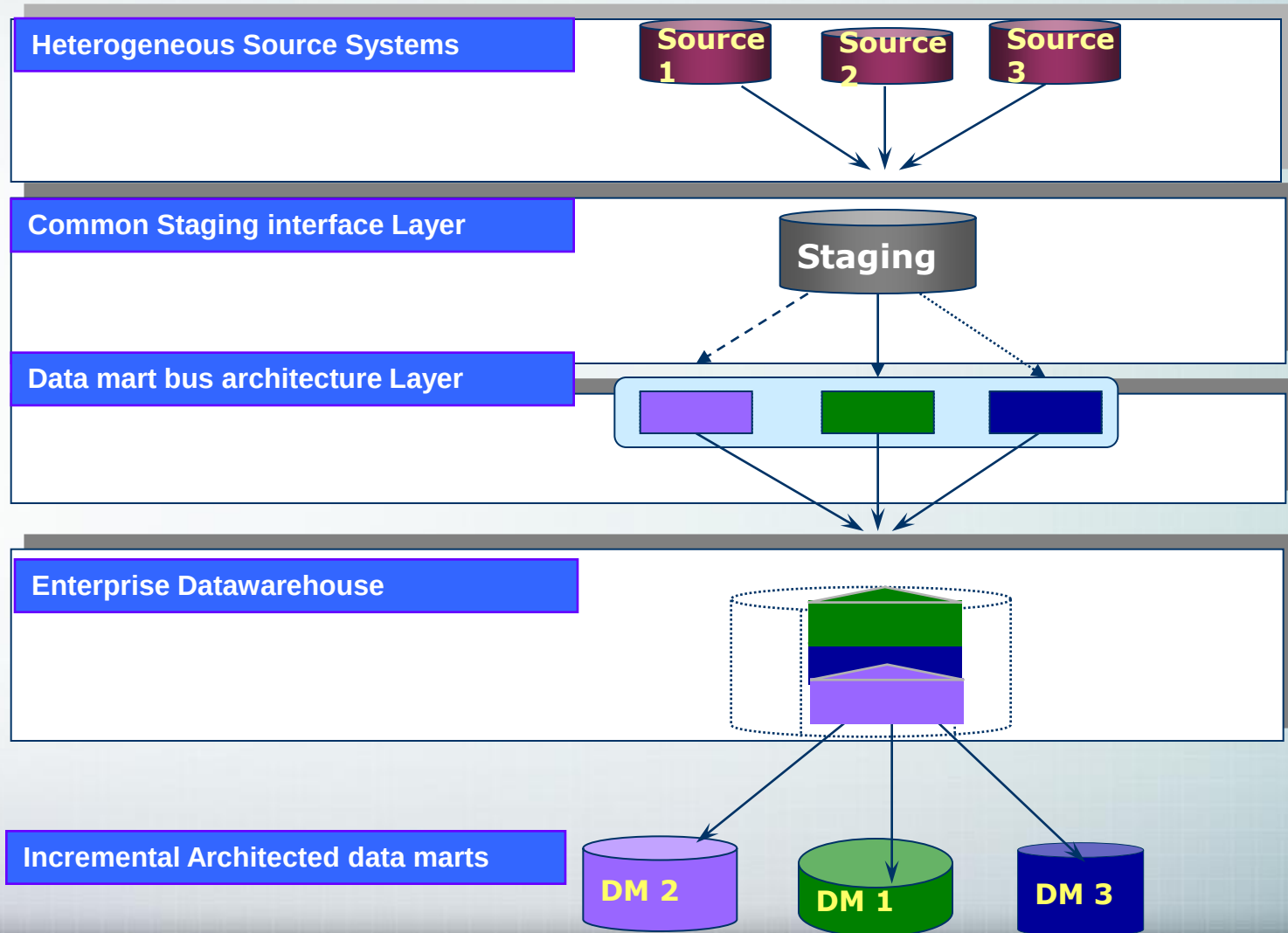
DWH DESIGN APPROACHES

- **Data warehouse design** is one of the key technique in building the data warehouse.
- Choosing a right data warehouse design **can save the project time and cost.**
- Basically there are two data warehouse design approaches are popular.

Top-Down Design:

- **In the top-down design approach the, data warehouse is built first.**
- **The data marts are then created from the data warehouse.**

"Top Down "Approach – Innmon's



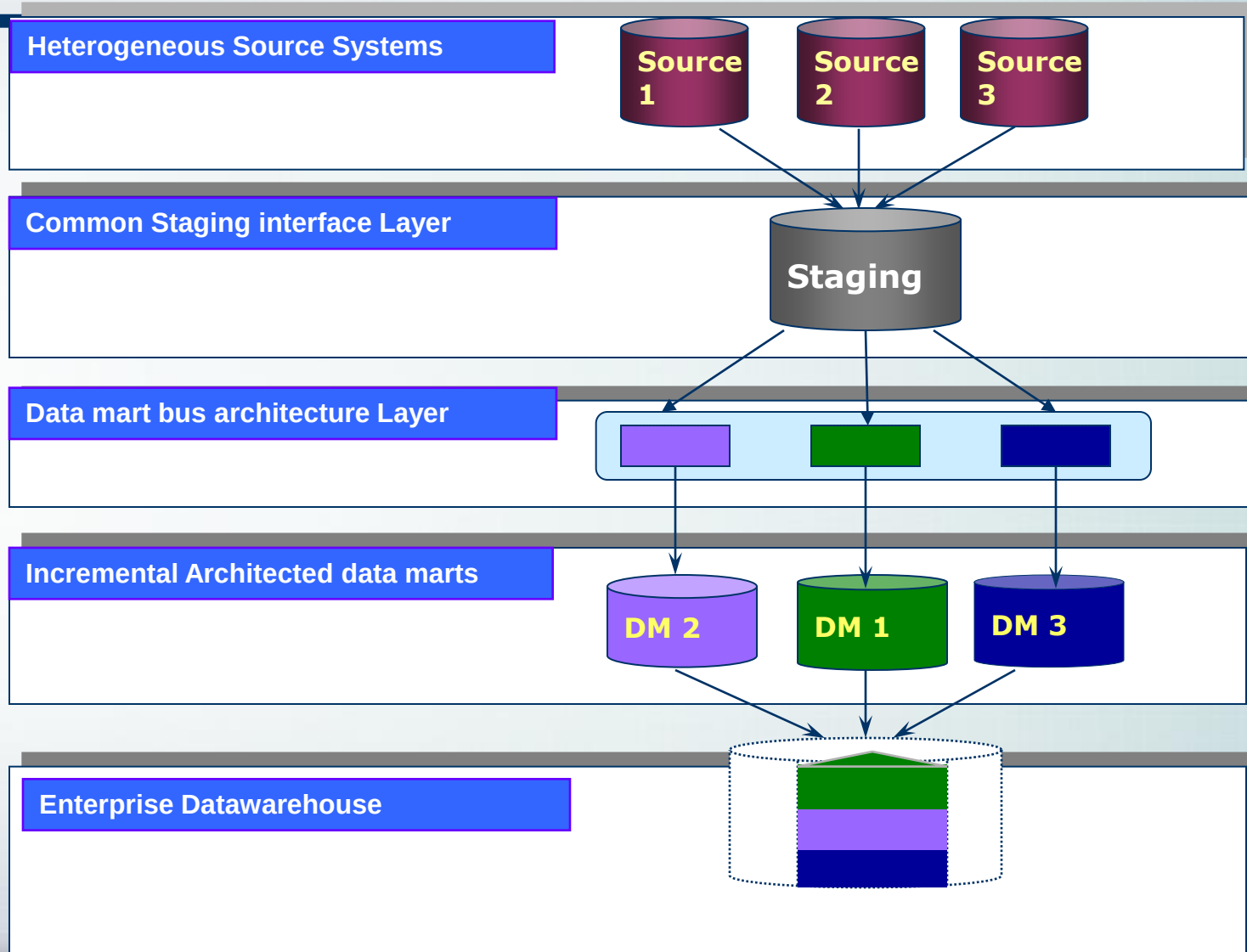
➤ **Advantages of top-down design are:**

- **Provides consistent dimensional views of data across data marts, as all data marts are loaded from the data warehouse.**
- **This approach is robust against business changes. Creating a new data mart from the data warehouse is very easy.**

Bottom-Up Design:

- **In the bottom-up design approach, the data marts are created first to provide reporting capability.**
- **A data mart addresses a single business area such as sales, Finance etc.**
- **These data marts are then integrated to build a complete data warehouse.**

“Bottom up “Approach – Kimball’s



Advantages of bottom-up design are:

- This model contains consistent data marts and these data marts can be delivered quickly.
- As the data marts are created first, reports can be generated quickly.
- The data warehouse can be extended easily to accommodate new business units. It is just creating new data marts and then integrating with other data marts.

OLTP Vs. ODS Vs. DWH

Characteristic	OLTP	ODS	Data Warehouse
Data redundancy	Non-redundant within system; Unmanaged redundancy among systems	Somewhat redundant with operational databases	Managed redundancy
Data stability	Dynamic	Somewhat dynamic	Static
Data update	Field by field	Field by field	Controlled batch
Data usage	Highly structured, repetitive	Somewhat structured, some analytical	Highly unstructured, heuristic or analytical
Database size	Moderate	Moderate	Large to very large
Database structure stability	Stable	Somewhat stable	Dynamic